Sustainable Hydrothermal Carbonization Synthesis of Iron/Nitrogen-Doped Carbon Nanofiber Aerogels as Electrocatalysts for Oxygen Reduction

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It is urgent to develop new kinds of low-cost and high-performance nonprecious metal (NPM) catalysts as alternatives to Pt-based catalysts for oxygen reduction reaction (ORR) in fuel cells and metal–air batteries, which have been proved to be efficient to meet the challenge of increase of global energy demand and CO_2 emissions. Here, an economical and sustainable method is developed for the synthesis of Fe, N codoped carbon nanofibers (Fe–N/CNFs) aerogels as efficient NPM catalysts for ORR via a mild template-directed hydrothermal carbonization (HTC) process, where cost-effective biomass-derived D(+)-glucosamine hydrochloride and ferrous gluconate are used as precursors and recyclable ultrathin tellurium nanowires are used as templates. The prepared Fe/N-CNFs catalysts display outstanding ORR activity, i.e., onset potential of 0.88 V and half-wave potential of 0.78 V versus reversible hydrogen electrode in an alkaline medium, which is highly comparable to that of commercial Pt/C (20 wt% Pt) catalyst. Furthermore, the Fe/N-CNFs catalysts exhibit superior long-term stability and better tolerance to the methanol crossover effect than the Pt/C catalyst in both alkaline and acidic electrolytes. This work suggests the great promise of developing new families of NPM ORR catalysts by the economical and sustainable HTC process.

1. Introduction

Development of advanced technical devices for energy storage and conversion, such as fuel cells and metal–air batteries, has been considered as an efficient strategy to meet the scientific challenges of growing energy demand and increasing CO_2 emissions over the world.[1–4] In fuel cells or metal–air batteries, oxygen reduction reaction (ORR) is unalterably occurred at the cathode and currently catalyzed by Pt-based electrocatalysts.[3,4] The commercialization of these energy devices is severely hampered compared to the gasoline engine, because of the prohibitive cost and scarcity of Pt-based catalysts.[5,6] Therefore, the design and synthesis of highly active and stable but less expensive metal-free or nonprecious metal (NPM) ORR catalysts that exclusively consist of earth-abundant elements, as alternatives to the Pt-based catalysts, have attracted great interests in the past several decades.[7–9]

It has been more than 50 years since Jasinski discovered for the first time that cobalt phthalocyanine had the potential in catalyzing the oxygen reduction in an alkaline electrolyte.[10] Later, several research groups demonstrated that other Co- and Fe–N_4 macrocycles were also active for the ORR in both alkaline and acidic media, even though the

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activity and the stability of these molecular catalysts were not good enough for practical applications.[11–14] A remarkable breakthrough work was done in 1981 by van Veen and Colijn, who found that heat-treatment of metal–N_4 macrocycles between 600 and 900 °C in an inert atmosphere could greatly enhance the ORR activity and stability in both alkaline and acidic media.[15] Another significant achievement was discovered in 1989, on which Yeager and co-workers found that the uneconomical macrocycles could be replaced by cheap, individual C, N, and metal precursors for the synthesis of metal/N/C catalysts (Me/N/C catalysts, Me is Fe and/or Co) via a pyrolysis process.[16] After that, a large number of Me/N/C catalysts were prepared by simultaneous heat-treating precursors of nitrogen, carbon supports and transition-metal salts at high temperature (600–1000 °C) in an reactive (NH_3) or inert (Ar or N_2) atmosphere.[17–21] Until now, the Me/N/C catalysts represent the most promising NPM ORR catalysts for fuel cells and metal–air batteries.[7,8,22,23]

Based on the pioneering works, it could be concluded that, to make the Me/N/C catalysts competitive, two critical factors should be considered to improve the ORR performance: (1) a high turnover frequency (TOF) which is considered as intrinsic property of the catalytic sites, and (2) a great density of accessible active sites.[24–27] In terms of the nature of active sites, it has been widely accepted that the highly active centers of Me/N/C catalysts consist of Fe (or Co) cations coordinated by pyridinic or pyrrolic N functionalities attached to carbon support.[28–30] The number of accessible active centers could be improved by carefully selecting precursors and synthetic parameters and controlling the porous structures.[27,31,32] The traditional method for preparing Me/N/C catalysts is based on the direct pyrolysis of the mixtures of carbon supports and nitrogen/metal precursors, which normally fails in controlling the porous structures, while the hard-templating synthesis always involves complex preparation processes.[26,33]

Besides the well-developed Me/N/C catalysts, independent efforts have been devoted recently to develop another promising type of carbon-based non-platinum ORR catalysts, i.e., nitrogen-doped carbons as metal-free catalysts.[9,34] It is believed that nitrogen-doping induces the charge delocalization and creates net positive charge on adjacent carbon atoms, which is favorable for O_2 adsorption to facilitate the ORR process.[35] Although the development of metal-free catalysts is fundamentally important for the understanding of ORR mechanism,[9,36] it has been widely accepted that introduction of some transition metals (e.g., Fe, Co) to the metal-free N-doped carbon frameworks results in metal–N_x active centers with much enhanced activity and stability in both alkaline and acidic media.[8]

Herein, we report a new family of Me/N/C catalysts consisting of uniform Fe and N-doped carbon nanofibers (CNFs) by a cost-effective template-directed hydrothermal carbonization (HTC) process, which was developed well recently by us and others for preparing functional nanostructured carbon materials.[37–43] Our previous work demonstrated that HTC treatment of nitrogen-containing carbohydrate in the presence of ultrathin tellurium nanowires (TeNWs) could produce N-doped CNFs aerogels which were efficient for catalyzing oxygen reduction as a metal-free catalyst.[44] This work aims to extend the template-directed HTC synthesis for preparing metal/N-doped CNFs with improved ORR performance. Abundant and economical carbohydrate salts, e.g., D-(+)-glucosamine hydrochloride and ferrous gluconate, are employed as molecular precursors for the template-directed HTC synthesis to prepare the Fe/N-containing carbonaceous nanofibers aerogels. Benefiting from the homogeneous distribution of iron/nitrogen precursors on the molecular level in the carbonaceous matrix, further pyrolysis and leaching treatment of the HTC products results in Fe/N-doped carbon nanofibers catalysts (Fe/N-CNFs) with abundant molecular FeN_x active sites and ideal porous structures. Consequently, the Fe/N-CNFs catalysts display comparable ORR activity to commercial Pt/C catalyst in alkaline media, and acceptable ORR performance in acidic media. Moreover, considering the use of biomass-derived precursors and recyclability of the TeNWs templates,[45] the current method is very attractive in terms of sustainability and cost compared to traditional hard-templating synthesis.

2. Results and Discussion

The synthetic procedure of the Fe/N-CNFs catalysts is illustrated in **Figure 1**. In the first step of the synthesis, ultrathin TeNWs templates were homogeneously dispersed in the mixed solution of D-(+)-glucosamine hydrochloride and ferrous gluconate. During the HTC treatment of the above solution, glucosamine hydrochloride and ferrous gluconate underwent dehydration and polymerization to form a homogeneous carbonaceous coating around the TeNWs templates, which resulted in a well-defined 1D core/shell structure (Figure 1a). It is reasonable to speculate that the iron and nitrogen elements distribute homogeneously throughout the whole carbonaceous nanoshells at the atomic scale. Afterward, a kind of mechanically robust monolith was obtained and could be taken out directly from the Teflon reactor without damage (the inset in **Figure 2**a). After etching out the TeNWs templates with acidic H_2O_2 solution and subsequently supercritical-drying with CO_2, Fe/N-containing carbonaceous nanofibers (denoted as HTC-Fe/N/C) aerogels were successfully prepared (Figure 1b).

Low-magnification scanning electron microscopy (SEM) image of the HTC-Fe/N/C aerogel showed a highly porous network structure constituted by disordered nanofibers (Figure 2a). Further SEM observation indicated that the uniform nanofibers formed a lot of junctions with each other during the HTC process (Figure S1, Supporting Information), which were hypothesized as the reason for the formation of the mechanically stable nanofiber monolith. An advantage of the template-directed HTC method is easy to control the diameter of the carbonaceous nanofibers in a wide range of tens to hundreds of nanometers by tuning the synthetic parameters, including the ratio of TeNWs/precursors, the HTC time and temperature (Figure 2b–f). For example, when keeping the TeNWs/precursors ratio and HTC temperature unchanged (32 mg TeNWs/2.0 g glucosamine hydrochloride/1.0 g ferrous gluconate, 180 °C), prolonging the HTC

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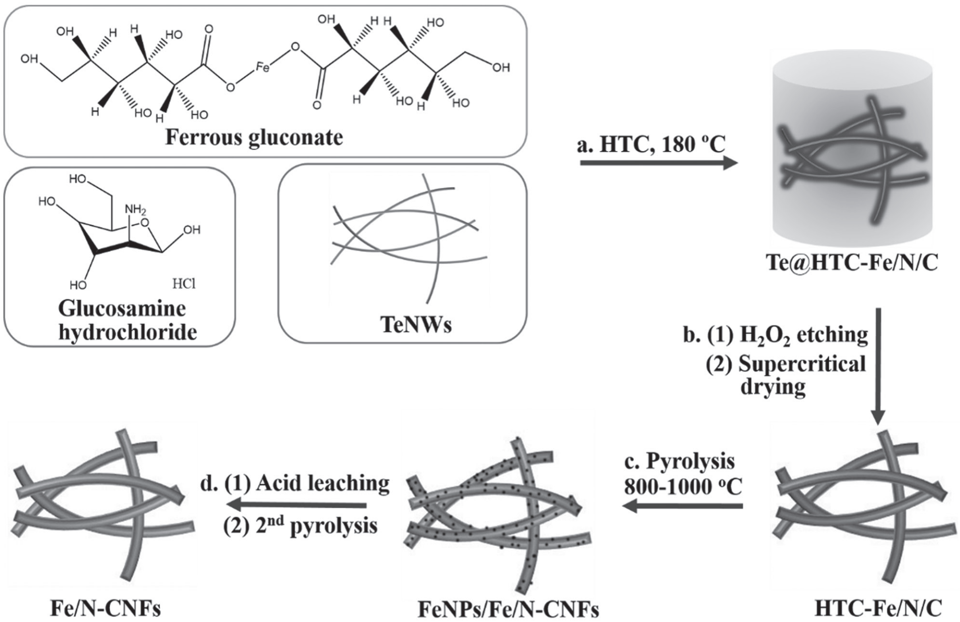

Figure 1. Schematic illustration of the synthesis of the Fe/N-CNFs catalysts. a) HTC treatment of a mixture of TeNWs, glucosamine hydrochloride, and ferrous gluconate at 180 °C created a monolithic wet gel composed of Te@Fe/N-containing carbonaceous nanocables. b) HTC-Fe/N/C aerogels were obtained after etching out the TeNWs templates and supercritical drying with CO_2. c) The HTC-Fe/N/C aerogels were pyrolyzed at high temperature under Ar atmosphere for the first time to prepare the FeNPs/Fe/N-CNFs. d) The final Fe/N-CNFs catalysts were prepared after acid leaching and the second pyrolysis treatment.

reaction duration from 10 to 36 h led to an increase of the diameter of carbonaceous nanofibers from ≈100 to 250 nm (Figure 2b,c,f). Meanwhile, decreasing the ratio of TeNWs/precursors would enable each TeNW template to receive more HTC precursors and finally result in the increased size of carbonaceous nanofibers (Figure 2d–f). Interestingly, the N-containing carbonaceous nanofibers (HTC-N/C) prepared without use of iron salt under the same conditions exhibited the similar diameter as the HTC-Fe/N/C nanofibers (Figure S2, Supporting Information), although it has been reported that metal ion could accelerate the formation of carbonaceous microspheres.[46]

The key point for the successful template-directed HTC synthesis is the selection of an organic Fe (II) salts (i.e., ferrous gluconate in the current work) as iron source. The use of inorganic Fe (III) or Fe (II) salts failed in the synthesis of uniform HTC-Fe/N/C (Figure S3, Supporting Information). Inorganic Fe (III) salts (i.e., $FeCl_3$) could oxidize the ultrathin TeNWs completely within 10 min, while Fe (II) salts (i.e., $FeSO_4$) caused serious aggregation of TeNWs

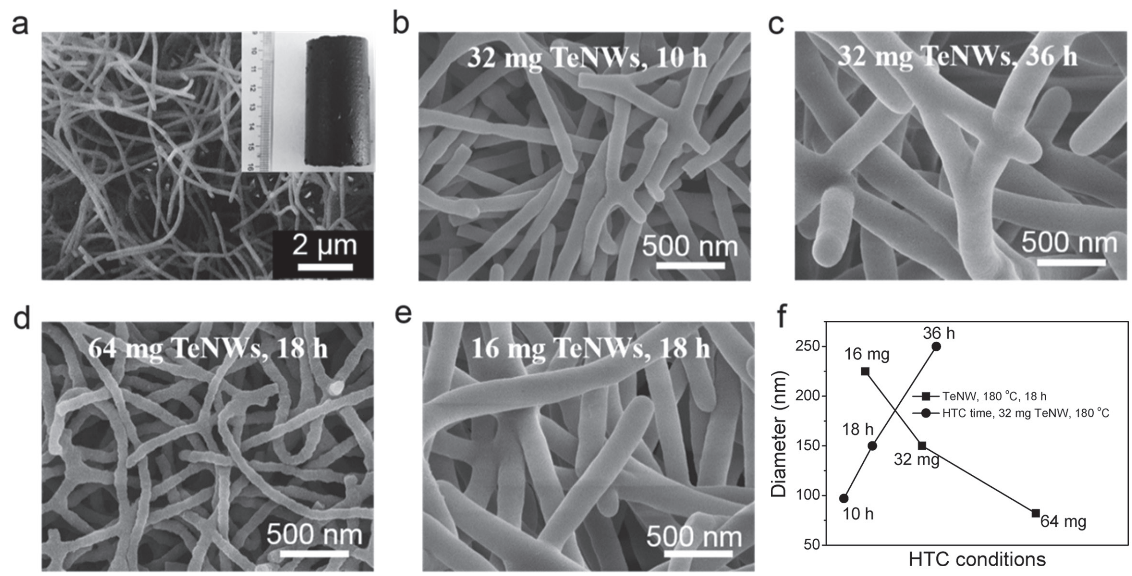

Figure 2. a) SEM image of the typical HTC-Fe/N/C aerogel. The inset shows the digital photo of the HTC-Fe/N/C aerogel. b–e) SEM images of the HTC-Fe/N/C aerogels with different diameters prepared at different HTC conditions. f) Dependence of the diameters of HTC-Fe/N/C nanofiber versus HTC reaction time or TeNWs amount.

(Figure S4, Supporting Information). In contrast, ultrathin TeNWs could be homogeneously dispersed in the ferrous gluconate or ammonium ferric citrate solution for a long time (Figure S4, Supporting Information). As expected, under the same HTC reaction conditions, replacement of ferrous gluconate with ammonium ferric citrate could produce uniform HTC-Fe/N/C as well (Figure S5, Supporting Information).

To fabricate the Fe/N-CNFs catalysts, the as-prepared HTC-Fe/N/C with a diameter of ≈150 nm were selected as the typical sample and were first pyrolyzed at 800–1000 °C under argon atmosphere to form a composite consisting of Fe-based inorganic nanoparticles and Fe/N-doped CNFs (denoted as FeNPs/Fe/N-CNFs) (Figure 1c). The FeNPs/Fe/N-CNFs samples were then treated with 0.5 м H_2SO_4 solution under 80 °C for 8 h to leach out the inorganic nanoparticles. Finally, an additional heat treatment was carried out after the leaching step to yield the final Fe/N-doped carbon nanofiber (denoted as Fe/N-CNFs) catalysts (Figure 1d). For comparison, metal-free N-doped CNFs (denoted as N-CNFs) catalysts were also prepared under similar conditions without using ferrous gluconate, as reported by our group previously.[44]

Transmission electron microscopy (TEM) image of the FeNPs/Fe/N-CNFs showed the presence of some irregular inorganic nanoparticles on the CNFs (**Figures 3**a). High-angle annular dark-field scanning TEM (HAADF-STEM) observation further confirmed the nanoparticle-carbon composite structure (Figure 3b). High-resolution TEM (HRTEM) indicated that the spacing of crystalline lattices of the inorganic nanoparticle in two directions were 0.169 and 0.198 nm, respectively (Figure S6, Supporting Information), which matches well with the (311) and (220) planes of the magnetism-Fe_2O_3 phases. After leaching by 0.5 м H_2SO_4, we could hardly observe any inorganic nanoparticles left in the Fe/N-CNFs catalyst (Figure 3c). HRTEM image displayed numerous mesopores on the Fe/N-CNFs catalyst (Figure 3d). Additionally, the randomly orientated graphene layers could be easily observed at the pore walls with interlayer spacing of 0.348 nm, which corresponds to the (002) plane of graphitic carbon. Importantly, elemental mapping with sub-nanoscale energy filtered TEM (EFTEM) imaging of Fe/N-CNFs revealed that both iron and nitrogen elements were homogeneously distributed throughout the whole carbon matrix after leaching out the inorganic Fe-containing nanoparticles (Figure 3e).The X-ray diffraction (XRD) pattern (**Figure 4**a) further identified that the inorganic nanoparticles in the FeNPs/Fe/N-CNFs had been removed out completely by the acid leaching.

To analyze the elemental content and the doped nitrogen functionality of the catalysts, Fourier transform infrared spectroscopy (FT-IR) and X-ray photoelectron spectroscopy (XPS) measurements were carried out to characterize the HTC-Fe/N/C, FeNPs/Fe/N-CNFs, and Fe/N-CNFs samples. As revealed by FT-IR, a large number of amino groups ($-NH_2$) located on the HTC-Fe/N/C, while almost all functional groups disappeared after the pyrolysis at high temperature (Figure S7, Supporting Information). The XPS analyses indicated that, during the first pyrolysis, the nitrogen content decreased from 7.51 at% for the HTC-Fe/N/C to 2.67 at% for the FeNPs/Fe/N-CNFs and the oxygen content decreased greatly from 19.29 to 3.38 at%. After acid leaching and second pyrolysis, the nitrogen content further decreased to 1.76 at% for the Fe/N-CNFs catalyst (Figure 4b and Table S1, Supporting Information). The significant decrease of nitrogen and oxygen contents is attributed to the thermal decomposition of amino and oxygen-containing groups at high temperature. The left nitrogen atoms in the Fe/N-CNFs catalyst should be doped into the graphene lattice of carbon (see below for more discussion).

No obvious Fe signals could be observed for all samples by XPS analysis because of the relatively high detection limit of the XPS technique. Therefore, inductively coupled plasma atomic emission spectrometry (ICP-AES) was taken

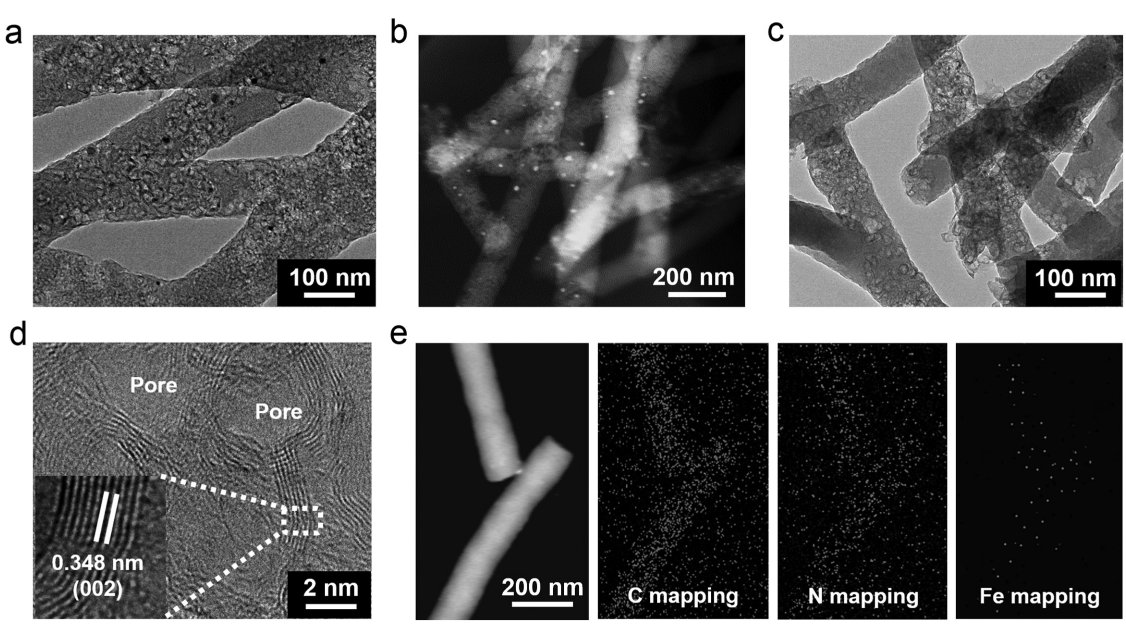

Figure 3. a,b) TEM and HAADF-STEM images of the FeNPs/Fe/N-CNFs, respectively. c,d) TEM and HRTEM images of Fe/N-CNFs, respectively. The inset in (d) shows the enlarged HRTEM image. e) EFTEM element mapping of the Fe/N-CNFs catalyst.

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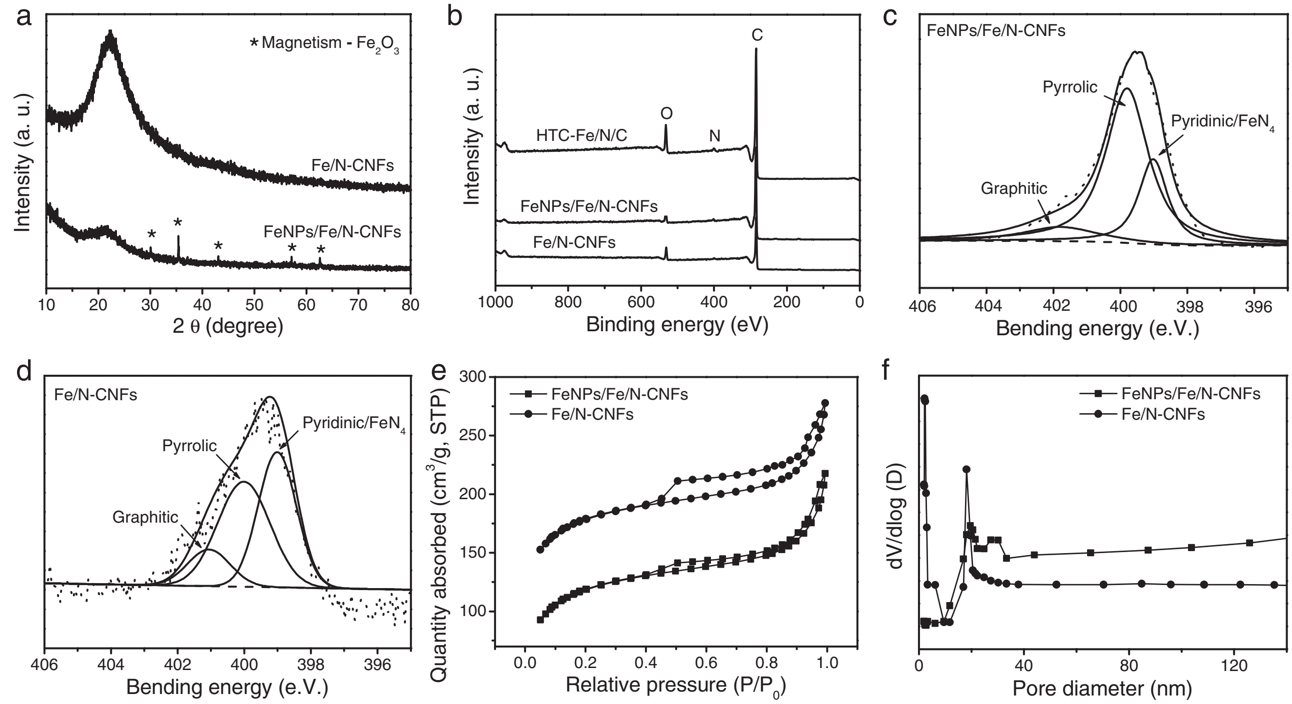

Figure 4. a) XRD patterns of FeNPs/Fe/N-CNFs and Fe/N-CNFs. b) XPS survey spectra of HTC-Fe/N/C, FeNPs/Fe/N-CNFs and Fe/N-CNFs. c,d) High-resolution XPS N 1s spectra of FeNPs/Fe/N-CNFs and Fe/N-CNFs, respectively. e,f) N_2 sorption isotherms and pore size distribution curves of FeNPs/Fe/N-CNFs and Fe/N-CNFs, respectively.

to analyze the change of Fe content during the pyrolysis and acid leaching. After the first pyrolysis, the Fe content slightly increased from 1.16 wt% for HTC-Fe/N/C to 1.3 wt% for FeNPs/Fe/N-CNFs because of the loss of carbon, nitrogen, and oxygen elements. The acid leaching and second pyrolysis caused a remarkable decrease of Fe content to 0.44 wt%, due to the removal out of inorganic Fe-containing nanoparticles by acid leaching.

The high-resolution N 1s spectra of FeNPs/Fe/N-CNFs and Fe/N-CNFs catalysts were displayed in Figure 4c,d. These spectra could be fitted well with three primary peaks, which assigned to the pyridinic-like N (398.6 eV), pyrrolic-like N (400.3 eV), and quaternary-like N (401.2 eV), respectively.[17,32] Both pyridinic-N and pyrrolic-N atoms were bonded with two C atoms and resided at the edges between two crystallites graphene layers.[2,18] The pyridinic-like N coordination, which were proposed to the essential part of the Fe-N_x active sites, were widely reported to be responsible for the high activity and stability of the Me/N/C catalysts.[21,44,47] Notably, after acid leaching and the second pyrolysis, the relative content of pyridinic-like N increased from 21% for FeNPs/Fe/N-CNFs to 44% for Fe/N-CNFs (Figure 4c,d).

The texture properties of FeNPs/Fe/N-CNFs and Fe/N-CNFs were analyzed by N_2 sorption tests. The hysteresis loops of type-IV indicated the mesoporous structure of both FeNPs/Fe/N-CNFs and Fe/N-CNFs (Figure 4e), which was consistent with the TEM observations (Figure 3c,d). The Brunauer–Emmett–Teller (BET) surface area and pore volume of FeNPs/Fe/N-CNFs were 350 $m^2\ g^{-1}$ and 0.20 $cm^3\ g^{-1}$, respectively. After acid leaching and the second pyrolysis, the BET surface area and pore volume of Fe/N-CNFs increased to 584 $m^2\ g^{-1}$ and 0.35 $cm^3\ g^{-1}$, respectively. The pores size distribution curves further indicated mesoporous feature of the obtained Fe/N-CNFs catalysts (Figure 4f). The high BET surface area and mesoporous structure of Fe/N-CNFs determined the high exposure of active sites and the favorable transport properties of ORR-relevant species.[19]

The ORR performance of the Fe/N-CNFs catalysts were evaluated using rotating disk electrode (RDE) technique in 0.1 M KOH or 0.5 M H_2SO_4 at room temperature. We first studied the ORR activity of the Fe/N-CNFs catalysts in 0.1 M KOH as a function of the amount of iron precursors and the pyrolysis temperature in the range of 800–1000 °C. The best ORR performance was achieved when 1.0 g ferrous gluconate was used for the HTC synthesis and the optimal pyrolysis was 900 °C (Figure S8, Supporting Information and **Figure 5**a). Therefore, the Fe/N-CNFs catalysts discussed below were prepared with 1.0 g ferrous gluconate at the pyrolysis temperature of 900 °C unless otherwise specified. For comparison, metal-free N-CNFs and Pt/C (20 wt% Pt) catalysts were also measured under the same conditions. Figure 5b showed the linear sweep voltammetry (LSV) curves of N-CNFs-900, FeNPs/Fe/N-CNFs, Fe/N-CNFs, and 20% Pt/C catalysts. Obviously, the N-CNFs-900 catalyst exhibited the worst ORR activity with the onset potential of 0.67 V versus reversible hydrogen electrode (RHE). As expected, the FeNPs/Fe/N-CNFs exhibited a much enhanced ORR performance (onset potential of 0.75 V and half-wave potential of 0.69 V) compared to the N-CNFs catalyst, indicating the formation of the highly active Fe–N_x active sites for the FeNPs/Fe/N-CNFs. After the acid leaching and second pyrolysis, the ORR activity was further improved for the

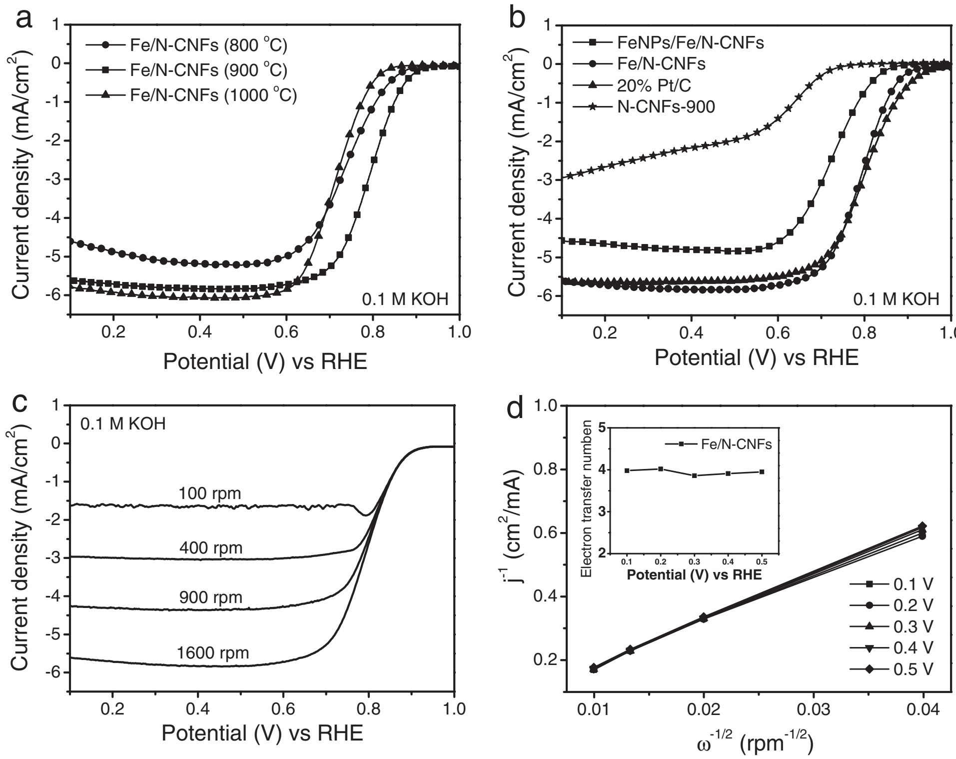

Figure 5. a) LSV curves of Fe/N-CNFs prepared at different pyrolysis temperature in an O_2-saturated 0.1 M KOH electrolyte. b) LSV curves of N-CNFs, FeNPs/Fe/N-CNFs, Fe/N-CNFs, and 20% Pt/C. c) LSV curves of Fe/N-CNFs at different rotating speeds from 100 to 1600 rpm. d) The corresponding Koutecky–Levich plots of Fe/N-CNFs from 0.1 to 0.5 V (vs RHE). The inset of (d) shows the electron-transfer number as function of the potential.

Fe/N-CNFs catalyst (onset potential of 0.88 V and half-wave potential of 0.79 V), which closely approach to that of Pt/C catalyst (onset potential of 0.95 V and half-wave potential of 0.80 V). The enhancement of ORR activity revealed that the acid leaching eliminated the inactive inorganic iron phase and exposed additional carbon surface, as confirmed by the N_2 sorption tests (Figure 4e), which in turn led to the enhanced ORR performance. Besides, the Fe/N-CNFs catalyst prepared from ammonium ferric citrate also exhibited high ORR activity (half-wave potential of 0.79 V), comparable to the activity of Fe/N-CNFs catalyst prepared from ferrous gluconate (Figure S9, Supporting Information), indicating that the precursors of organic Fe salts have negligible influence on the ORR performance. The selectivity of the four-electron reduction of oxygen for the Fe/N-CNFs catalyst was also studied using the RDE tests on at various rotating speeds (Figure 5c). The linear K–L plots (j^{-1} vs $\omega^{-1/2}$) displayed well linearity between the potential ranging from 0.1 to 0.5 V (Figure 5d). As displayed in the inset of Figure 5d, the electrons transfer number (n) of the Fe/N-CNFs catalyst was around 4, which means a favorable four-electron ORR process.

To demonstrate the potential of Fe/N-CNFs catalyst for ORR in acidic media, we also investigated the ORR performance of the Fe/N-CNFs catalyst in 0.5 M H_2SO_4. The Fe/N-CNFs catalyst displayed an onset potential of 0.67 V and a half-wave potential of 0.62 V, which were roughly 0.21 V lower than those of the commercial 20% Pt/C catalyst (Figure S10a, Supporting Information). RDE experiments at different rotating speeds (Figure S10b, Supporting Information) were carried out from 100 to 1600 rpm and analyzed by K–L equation (Figure S10c, Supporting Information).The electron transfer number was calculated to be very close to 4 from 0.1 to 0.5 V (Figure S10d, Supporting Information), indicating that a four-electron pathway dominated the ORR process in the acid electrolyte. Significantly, the chronoamperometric tests (Figure S11a,b, Supporting Information) showed that Fe/N-CNFs catalyst had a better long-term durability than Pt/C catalyst in both acid and basic electrolyte.

Furthermore, the performance of direct methanol fuel cells (DMFCs) also relies on the ORR at the cathode. Commercial 20% Pt/C catalysts suffered from indiscriminate nature of Pt-based catalysts, which catalyzes both the ORR and oxidation of crossover methanol at the DMFCs cathode at the same time.[48] Chronoamperometric responses of 1 M methanol introduced into alkaline or acidic electrolyte were tested for both Fe/N-CNFs and commercial 20% Pt/C to investigate the potential application of Fe/N-CNFs in DMFCs. Figure S11c,d displayed a drastic drop of instantaneous current of Pt/C with the addition of 1 M methanol in alkaline or acidic solution while the current of Fe/N-CNFs showed less changes. A possible reason of this undesirable activity loss for Pt/C catalyst might be owed to the initial methanol adsorption and dehydrogenation processes which were taken place on the Pt sites, leading to the formation of chemisorbed CO that strongly bound to catalyst surface.[48] Consequently, although Fe/N-CNFs displayed less active for ORR than 20% Pt/C in acid electrolyte, its high methanol tolerance also made Fe/N-CNFs catalyst attractive for DMFCs.

To further understand the nature of the active sites, we compared the ORR polarization curves of Fe/N-CNFs in the

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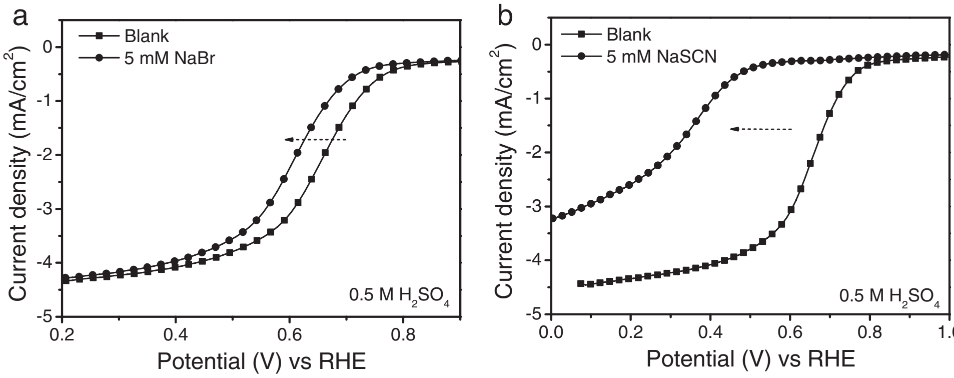

Figure 6. Effects of a) Br^- and b) SCN^- ions on ORR activities of the Fe/N-CNFs catalyst in 0.5 M H_2SO_4. The concentrations of both ions is 5×10^{-3} M.

0.5 M H_2SO_4solution before and after adding 5×10^{-3} M Br^- or SCN^- ions into the acidic electrolyte (**Figure 6**a,b).[49,50] Once introducing 5×10^{-3} M Br^-, the half-wave potential of Fe/N-CNFs was negatively shifted 24 mV and the limited diffusion current density nearly unchanged. However, the half-wave potential was significantly shifted 205 mV and the limited diffusion current density decreased by 40% at 0.2 V after adding 5×10^{-3} M SCN^- into the electrolyte. Clearly, Br^- ions could slightly harm the ORR performance while SCN^- ions seriously poisoned the activity sites. These results, combined with the XPS and EFTEM mapping analyses, strongly suggested that the ionic iron species were well distributed in the carbon/nitrogen matrix and coordinated with nitrogen to form active and stable FeN_x centers for ORR in the Fe/N-CNFs catalysts.[49]

3. Conclusion

In summary, we have demonstrated an economical and sustainable template-directed HTC process to prepare Fe, N codoped carbon nanofibers ORR catalysts, in which biomass-derived carbohydrate salts were used as precursors and recyclable ultrathin TeNWs were used as templates. The prepared Fe–N/CNFs catalysts exhibited outstanding ORR activity, excellent long-term stability, and tolerance to the methanol crossover effect in both alkaline and acid media. Benefiting from the robust and versatile HTC process, the current synthetic strategy would be extended to fabricate other transition metal-centered catalysts, e.g., Co/N-doped carbon (Figure S12, Supporting Information), bimetallic catalysts (e.g., FeCo–N/CNFs), even or other coordinated heteroatoms (e.g., sulfur), which could potentially lead to an enhancement in the ORR performance and finally enable the catalysts to compete with the Pt/C catalyst in technological devices.

4. Experimental Section

Chemicals: All the chemicals were analytical grade and commercially available from Shanghai Chemical Reagent Co. Ltd and were used without further purification.

Synthesis of Ultrathin Te Nanowire Templates: The ultrathin TeNWs were synthesized by a simple hydrothermal method reported previously by our group.[51] In a typical procedure, 10.0 g PVP (poly vinyl pyrrolidone) was dissolved in 320 mL of double distilled water under vigorous magnetic stirring to from a homogeneous solution at room temperature. Then, 922 mg sodium tellurite (4.16 mmol Na_2TeO_3) was added into the previous solution and dissolved before 16.7 mL hydrazine hydrate ($N_2H_4 \cdot H_2O$, 85%) and 33.3 mL aqueous ammonia ($NH_3 \cdot H_2O$, 25%–28%) were added into the mixture solution. The clear mixed solution was then transferred into a 500 mL Teflon-lined stainless steel autoclave. Finally, the autoclave was sealed and maintained at 180 °C for 3 h in a digital-temperature-controlled oven and then allowed to cool to room-temperature rapidly by water.

Synthesis of Highly Uniform HTC-Fe/N/C Aerogels: In a typical experiment for preparation of HTC-Fe/N/C nanofiber with a diameter of ≈150 nm, 200 mL of acetone was added into 60 mL of TeNWs solution to precipitate the products, which was collected by centrifuged at 8000 rpm for 3 min, and then washed several times with double-distilled water and absolute ethanol. Then, ≈64 mg ultrathin TeNWs were achieved. Afterward, the as-prepared ultrathin TeNWs were dispersed into 35 mL mixed solution consisting of 2.0 g D(+)-glucosamine hydrochloride (>99.0%) and 1.0 g ferrous gluconate (>99%) with vigorous magnetic stirring for 15 min. Hydrothermal treatment of the mixed solution at 180 °C for 18 h using an autoclave with a total volume of 50 mL resulted in a gel-like product, which was then taken out from the Teflon container directly and soaked in acidic H_2O_2 solution ($HCl:H_2O_2:H_2O$ = 1:2:10, v/v, 5 h), distilled water (12 h × 5 times) and ethanol (12 h × 5 times) successively at room temperature to remove the impurities and TeNWs templates. Finally, the water in the purified gels was exchanged with acetone and then the gels was supercritical-dried with CO_2 to obtain the HTC-Fe/N/C aerogels.

Synthesis of Fe/N–CNFs Electrocatalyst: The as-prepared HTC-Fe/N/C aerogels were first carbonized under a flowing inert atmosphere (Ar) at the temperature ranging from 800 to 1000 °C for 2 h to yield the FeNPs/Fe/N–CNFs sample. To remove the iron-containing inorganic nanoparticles that were considered as the unstable and ORR-unreactive phases, the FeNPs/Fe/N–CNFs sample was then leached in 0.5 M H_2SO_4 at 80 °C for 8 h, followed by filtration collection, thoroughly washing with deionized water to neutral and dried. Finally, the leached sample was pyrolyzed again in Ar atmosphere at the same temperature for 2 h. For comparison, a metal-free nitrogen-doped carbon nanofibers (N-CNFs) were prepared by the same procedure without using ferrous gluconate, as reported previously by our group.[44]

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Characterization: SEM was performed on a Zeiss Supra 40 scanning electron microscope at an acceleration voltage of 5 kV. TEM images were acquired on a Hitachi H7650 transmission electron microscope with a CCD imaging system and an accelerating voltage of 120 kV. HRTEM and EFTEM mapping were performed using a JEM-ARM 200F microscope operating at an accelerating voltage of 200 kV. Elemental mapping were collected using a Gatan GIF Quantum 965. Inductively coupled plasma atomic emission spectrometry measurements (ICP-AES) were conducted on an Atomscan Advantage Spectrometer (Thermo Ash Jarrell Corporation). N_2 sorption analysis was conducted using a TriStar 3020 accelerated surface area and porosimetry instrument (Micromeritics), equipped with automated surface area, at 77 K using BET calculations for the surface area. The pore size distribution plot was recorded from the adsorption branch of the isotherm based on the Barrett–Joyner–Halenda (BJH) model. X-ray photoelectron spectra (XPS) were recorded on an ESCALab MKII X-ray photoelectron spectrometer, using Mg Kα radiation as the excitation.

Electrocatalytic Measurements: The electrochemical measurements were carried out in a conventional three-electrode cell using IM6e electrochemical workstation (ZahnerElectrik, Germany) controlled at room temperature. A 5.0 mm diameter glassy carbon disk (disk geometric area 0.196 cm^2) served as the substrate for the working electrode, with the rotating rate varying from 100 to 1600 rpm. Ag/AgCl (3.5 M KCl) and Pt wire were used as reference and counter electrodes in alkaline electrolyte (0.1 M KOH), respectively. In acid electrolyte (0.5 M H_2SO_4), saturated calomel electrode (SCE) were taken as the reference electrode and Pt wire was also used as counter electrodes. The potential difference between reference electrode and RHE was determined based on the calibration measurement in H_2-saturated electrolytes. The Fe-based catalyst or commercial Pt/C (20 wt%, Johnson Matthey) catalyst ink was prepared by blending the catalyst powder (2 mg) with 20 µL Nafion solution and 980 µL ethanol by sonication. 20 µL of Fe-based catalyst ink or 10 µL of commercial 20% Pt/C was then pipetted onto the glassy carbon surface. Electrolyte was saturated with N_2/O_2 by bubbling N_2/O_2 prior to the start of each experiment for 30 min. The cyclic voltammetry (CV) tests were performed in N_2- and O_2-saturated aqueous electrolyte solutions with a scan rate of 50 mV s^{-1}. RDE tests were measured in O_2-saturated electrolyte with a sweep rate of 5 mV s^{-1}. In order to estimate the double layer capacitance, the electrolyte was deaerated by bubbling with N_2, and then the voltammogram was evaluated again in the deaerated electrolyte. The oxygen reduction current was taken as the difference between currents measured in the deaerated and O_2-saturated electrolytes.

Electron transfer numbers were calculated using the Koutecky–Levich equation

$$\frac{1}{j}=\frac{1}{j_L}+\frac{1}{j_K}=\frac{1}{B\omega^{1/2}}+\frac{1}{j_K} \tag{1}$$

$$B=0.62nFC_0(D_0)^{2/3}\nu^{-1/6} \tag{2}$$

$$j_K=nFkC_0 \tag{3}$$

where j is the measured current density, j_L and j_K are the diffusion-limiting and kinetic current densities, respectively. n represents the overall number of electrons gained per O_2, F is the Faraday constant (F = 96485 C mol^{-1}), C_0 is the bulk concentration of O_2 (1.2×10^{-6} mol cm^{-3}, both 0.1 M KOH and 0.5 M H_2SO_4), D_0 is the diffusion coefficient of O_2 in electrolyte (1.9×10^{-5} cm^2 s^{-1}, both 0.1 M KOH and 0.5 M H_2SO_4), ν is the kinetic viscosity of the electrolyte (0.01 cm^2 s^{-1}, both 0.1 M KOH and 0.5 M H_2SO_4), ω is the angular velocity of the disk ($\omega = 2\pi N$, N is the linear rotation speed), and k is the electron transfer rate constant.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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